OPTIMIZATION OF citi bike STATION OCCUPANCY

Spencer Arnold · Paris Phan · Garrett Uthlaut

SYS 3060 — Stochastic Decision Models · University of Virginia · Spring 2026

THE PROBLEM

Citi Bike is the nation's largest bikeshare program, with 25,000 bikes and over 1,500 stations across Manhattan, Brooklyn, Queens, the Bronx, Jersey City, and Hoboken.

However, data suggests that these station placements and capacity do not adequately account for areas of greater attraction. As a result, docks in dense, high-traffic areas are often empty while other docks are completely full.

These imbalances cause **two forms of failure**:

- **Stockout**: riders are unable to access bikes in high-demand locations
- **Dockblock**: riders are unable to conveniently return bikes near their destinations in saturated areas.

This project looks to model and quantify these dynamics using stochastic processes, with the goal of evaluating strategies to improve performance.

RESEARCH QUESTION

Given a fixed total rebalancing budget $\Theta_{total} = \sum_{\mathrm{n}}(\theta_{\mathrm{n}})$ across the cluster,

what allocation $(\theta_1, \theta_2, \dots, \theta_n)$ minimizes the total long-run failure rate?

And how does the achievable minimum vary with the budget Θ_{total}?

If we introduce rebalancing trucks, how often should each station get visited, and how much rebalancing is enough before more stops helping?

ASSUMPTIONS

Demand

- Inter-arrival times are exponential within the 7–10 AM window
- Rental rate λ_n and return rate μ_n are constant across the window
- Users who find a station empty or full exit (no balking to neighbors)

Rebalancing

- Each station's rebalancing is modelled as an independent Poisson process at rate θ_n (not a shared, routed fleet)
- Rebalancing events reset state instantaneously to the target $t_n = \lfloor c_n/2 \rfloor$
- The truck fleet is assumed sized to support any $(\theta_1, \ldots, \theta_n)$ allocation and we optimize the *demand* side of rebalancing, not vehicle routing

DATA COLLECTION

For this project, we are using the data set "202109-citibike-tripdata". This contains bike rental data in December 2021. The data includes:

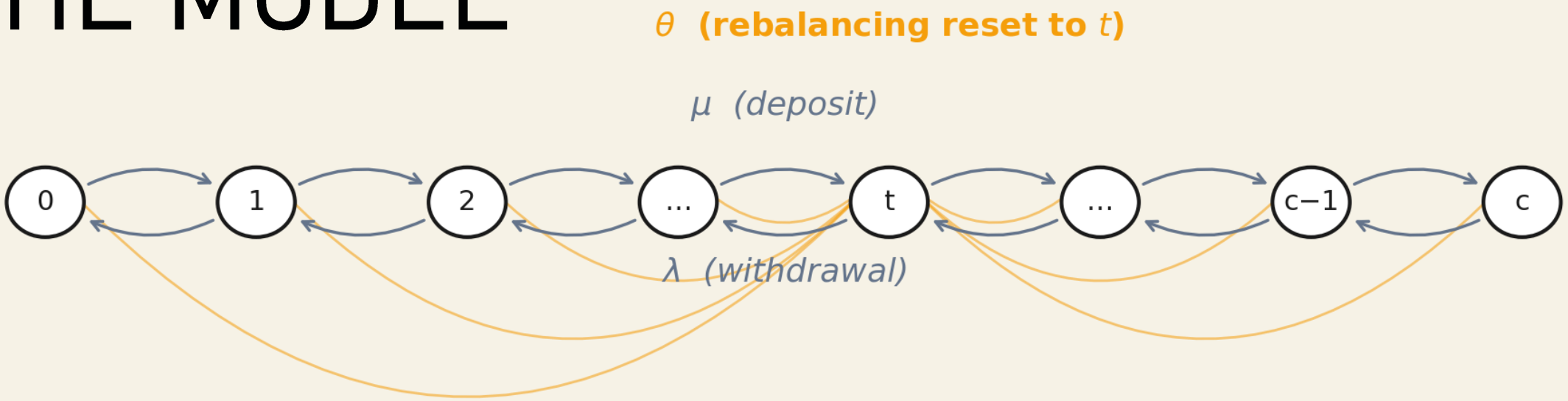
- Ride ID
- Rideable type
- **Started at**
- **Ended at**
- **Start station name**
- Start station ID
- **End station name**
- End station ID
- Start latitude
- Start longitude
- End latitude
- End Longitude
- Member or casual ride

station_id	name	capacity	λ	μ	ρ̂	log_imbalance	n_events	direction
8841.03	W Mosholu Pkwy S & Sedgwick Ave	37	0.048	1.873	39.333	+3.672	121	sink
7152.10	24 Ave & 26 St	19	2.698	0.079	0.029	-3.526	175	source
4804.02	Dock 72 Way & Market St	22	1.016	26.746	26.328	+3.271	1749	sink
3460.01	2 Ave & 36 St	43	0.111	2.794	25.143	+3.225	183	sink
4843.01	5 St & Market St	39	0.460	11.365	24.690	+3.206	745	sink
7188.10	E 81 St & Park Ave	46	6.810	0.317	0.047	-3.066	449	source
7932.10	W 144 St & Adam Clayton Powell Blvd	24	2.270	0.127	0.056	-2.883	151	source
5241.02	Metropolitan Ave & Vandervoort Ave	24	0.111	1.841	16.571	+2.808	123	sink
8738.01	Goulden Ave & Bedford Park Blvd	19	0.095	1.556	16.333	+2.793	104	sink
4920.13	South St & Broad St	31	0.127	2.032	16.000	+2.773	136	sink

Figure: Parsed data surfacing optimal station clusters for modelling

With this data, we can determine the average occupancy rate at docks throughout NYC.

THE MODEL



State $s_n \in \{0, 1, \ldots, c_n\}$ = number of bikes currently at station n. Target level $t_n = \lfloor c_n / 2 \rfloor$ (trucks restore to half-full).

TRANSITIONS

μ **Deposit (k → k+1)**
User returns a bike

θ **Rebalancing reset (k → t_n)**
Truck restores fill level

λ **Withdrawal (k → k−1)**
User takes a bike

Per-Station Failure Rate

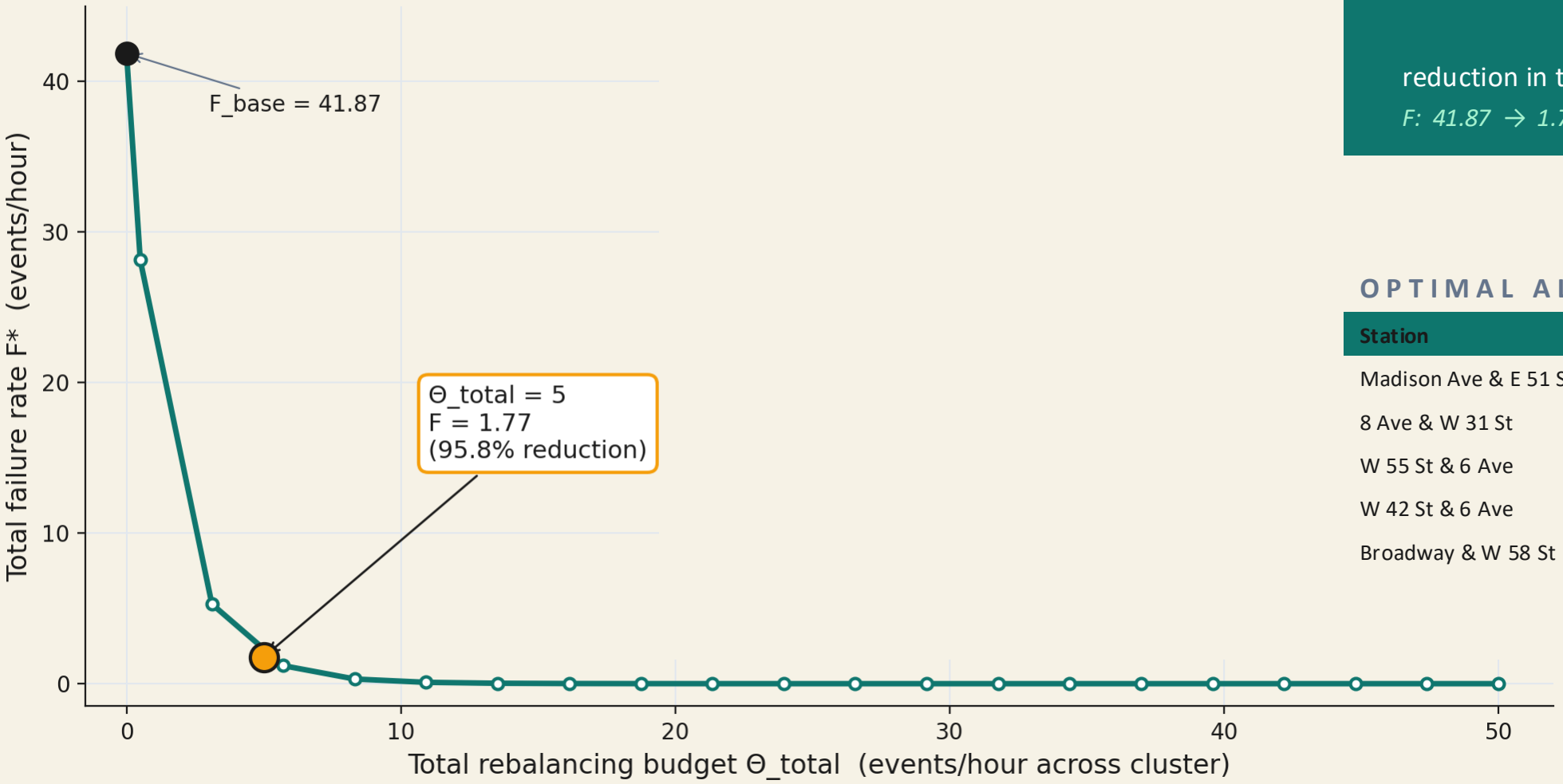
$$F_n(\theta_n) = \lambda_n \cdot \pi_0(n;\, \theta_n) + \mu_n \cdot \pi_{c_n}(n;\, \theta_n)$$

Stationary distribution π solves πQ = 0 as an exact linear system

Markov chain image created with AI

ANALYSIS & FINDINGS

Pareto frontier: cluster failure rate vs. rebalancing budget



AT Θ_total = 5 events/hour

95.8%

reduction in total failure rate

F: 41.87 → 1.77 events/hour

OPTIMAL ALLOCATION

Station	θ_n*	% drop
Madison Ave & E 51 St	**2.42**	94.8%
8 Ave & W 31 St	**1.09**	97.5%
W 55 St & 6 Ave	**0.79**	96.8%
W 42 St & 6 Ave	**0.59**	93.8%
Broadway & W 58 St	**0.11**	87.1%

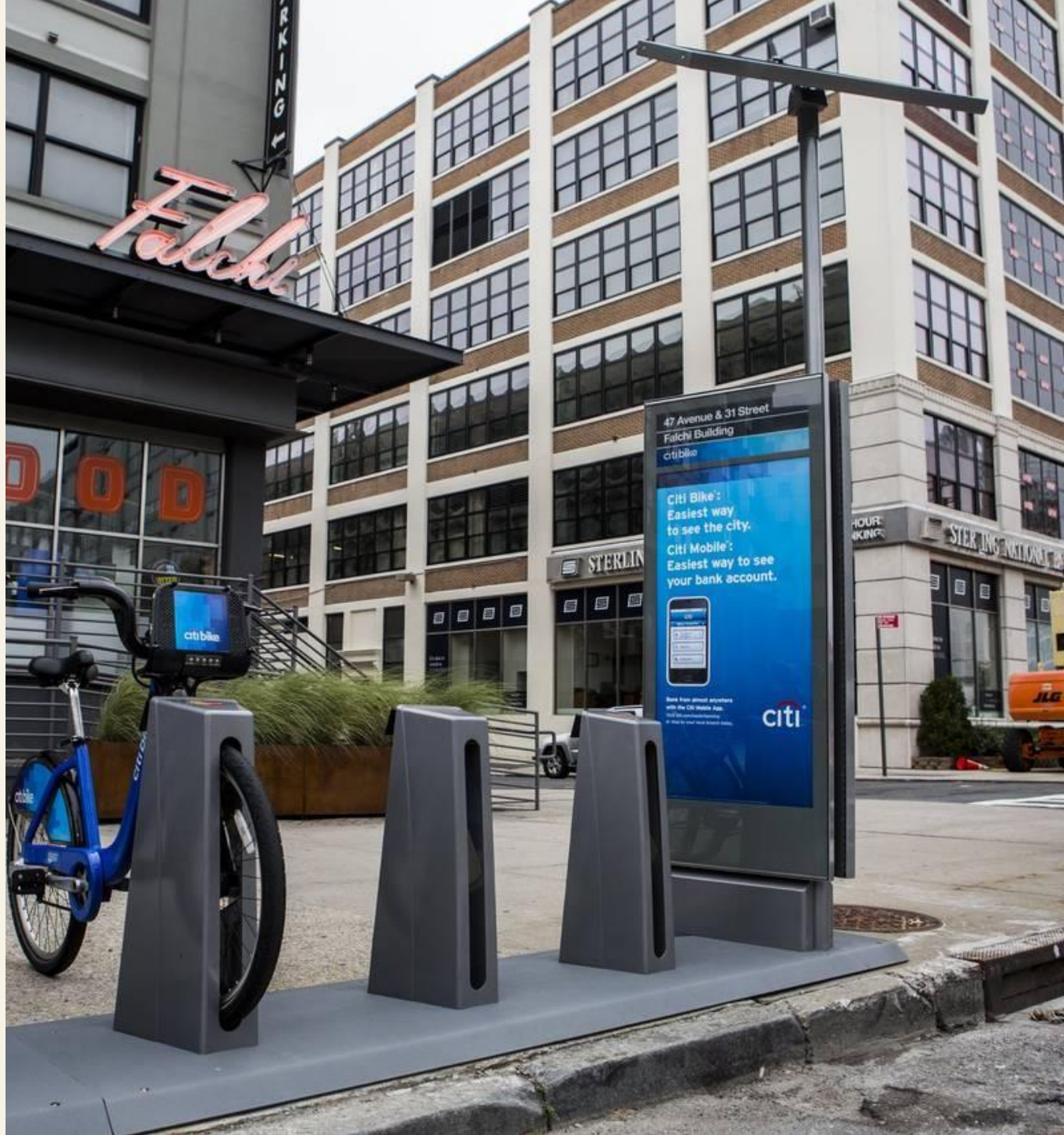
Graph created with AI

FUTURE IMPROVEMENTS: RIDER DESTINATION INCENTIVES

One potential approach to improving dock usage efficiency is implementing price adjustments based on where a user rents or returns a bike.

Prices could adjust to reflect real-time demand at each station, showing locations that either need more bikes or have excess capacity.

For example,

- Riders receive a discount for returning a bike to a further, more optimal station
- Incentives offered to riders for renting bikes from underutilized stations

This encourages users to naturally redistribute bikes across the network, turning them into a proxy relocation vehicle.